

# Daniel Hardesty Lewis

*Staff ML Engineer – production training, inference, evaluation, reliability*

Specializing in automated validation frameworks and scalable distributed infrastructure. Delivered consumer pricing and statewide resiliency platforms across geospatial & time-series workloads.

## Core Expertise

- 8 years **ML Engineering**: Build and operate AI/ML training and inference pipelines, using rigorous validation protocols to ensure model generalization and reliability
- 8 years **Backend & Data**: Build schemas and ETL; ship API-backed services using Python, SQL, and Postgres/PostGIS
- 8 years **Distributed Systems**: Scale batch workloads across distributed systems, profile bottlenecks, and optimize latency & throughput with explicit cost-performance tradeoffs
- 10 years **Geospatial AI**: Spatiotemporal modeling for risk, planning, and hazard decision-support use cases
- 8 years **Technical Leadership**: Drive cross-functional execution via specifications and governance; mentor engineers through design reviews, code reviews, and instruction

## Technologies

|                |                                                                                                        |
|----------------|--------------------------------------------------------------------------------------------------------|
| Causal AI      | DoWhy; EconML; PyWhy; CausalNex; Double ML; Causal Forests                                             |
| Modeling       | PyTorch; TensorFlow; scikit-learn; Transformers; LoRA/QLoRA/PEFT                                       |
| LLM apps       | embeddings; RAG; LangChain; LangGraph; DSPy; FAISS                                                     |
| Data stack     | Python; SQL; Pandas; Polars; PyArrow; Parquet; Postgres; PostGIS; Redis; Elasticsearch; Supabase       |
| Infrastructure | Linux/Unix; Bash; Git; Docker; Kubernetes; Modal; GCS; Ray; Spark; Dask; Airflow; CI/CD; observability |
| Additional     | CUDA; C++; TypeScript; React                                                                           |

## Experience

### Professional

- Jun. 2026 – Present **Entrepreneur-in-Residence**, [South Park Commons](#), New York, NY  
Selective technology community for founders at a leading venture capital firm.
  - Designed and implemented the next iteration of Homecastr's machine learning architecture: a full-sequence inpainting editable world model.
  - Conducted customer discovery with potential ICPs, including Parcel, Crusoe, and Cushman & Wakefield, to align predictive capabilities with enterprise requirements.
- Sep. 2025 – Present **Founder**, [Homecastr](#), New York, NY  
Consumer AI forecasting platform serving thousands of monthly active users. Owned the end-to-end ML architecture from data ingestion to web delivery.
  - Architected a nationwide neighborhood-level diffusion forecasting model, **matching Zillow's 7.3% 0-year error benchmark at a 1-year forecast horizon** and maintaining a **35.6% median error over 4 years**.
  - Engineered architecture incorporating **spatial inducing tokens**, **per-horizon loss normalization**, and **DDIM sampling** to balance predictive precision with calibrated price trajectories.
  - Developed predictive validation optimizing PIT uniformity, interval coverage, and sharpness to **rigorously validate forecast reliability** before production delivery.
  - Owned the end-to-end pipeline from data ingestion to real-time serving and refined product strategy through direct feedback from **a16z** and **MetaProp**.
- Oct. 2024 – Sep. 2025 **Technical Advisor**, [PoliBOM](#), Seattle, WA  
AI tariff mitigation and trade compliance product for manufacturers, centered on agentic BOM analysis and policy scenario simulation.
  - Engineered an agentic retrieval-augmented pipeline for BOM parsing and tariff simulation, replacing manual trade compliance workflows with predictable multi-step reasoning.
  - Directed development of a **conversational AI interface** enabling decision-makers to instantly **retrieve BOMs and receive proactive compliance alerts**.
  - Executing customer discovery with ScoutyAI and Columbia economists, and secured early market validation by presenting the prototype to KPMG.
- Nov. 2023 – Aug. 2025 **Founder**, [Summit Geospatial](#), New York, NY  
High-resolution elevation data and web delivery for engineering and hazard workflows.
  - Engineered a **statewide seamless Texas DEM mosaic**, resampling 0.5 m LiDAR sources to 1.2 m via nearest-neighbor across **70+ state, federal, and local elevation datasets**.
  - Developed the web distribution platform end-to-end including **data pipeline, tiling, hosting, delivery UX** to support engineering and planning use cases.

- Aug. 2021 – **Senior Data Scientist & Technical Lead**, [Texas Advanced Computing Center \(TACC\)](#), Austin, TX
- Aug. 2023
- Spun out Summit Geospatial alongside UT's VPs of IP to commercialize state-level terrain models from the \$40M [Texas Disaster Information System \(TDIS\)](#), delivering precision decision-support for civil engineering.
  - Developed inter-agency RFP and MOU frameworks with state and federal partners, translating technical requirements into scope and milestones facilitating [Real-Time Inundation Mapping](#) and TDIS.
  - Scaled climate and flood models on supercomputers, executing 800,000-node distributed jobs while managing \$1M+ compute budgets and federal partnerships with NOAA, NSF, USACE, GLO.
  - Developed efficient methods to produce high-resolution 1m flood maps from National Water Model outputs for Texas Emergency Management's Real-Time Inundation Mapping.
  - Partnered with the US Army Corps of Engineers to statistically model compound flood hazards in coastal Texas, integrating high-resolution topographic data with hydrological models.
  - Contributed research supporting Paola Passalacqua's [2022 EGU Bagnold Medal](#), one of geomorphology's highest honors.
  - Served as Technical Reviewer for [Frontera Doctoral Fellows](#) (2023) and [Large-Scale Computing Pathways](#) (2022) applications, evaluating \$10M+ in computing resource allocations on one of the world's largest supercomputers.
- Feb. 2018 – **Data Scientist & Research Engineer**, [Texas Advanced Computing Center \(TACC\)](#), Austin, TX
- Jul. 2021
- Competed in [DARPA World Modelers](#) to improve disaster resiliency and food security modeling in East Africa.
  - Mentored Petrobras engineers in deep learning and HPC to enable institutional technology transfer.
  - Cited by AAI President Yolanda Gil in her [farewell address](#) for contributions to automated scientific modeling.

## Research

- Sep. 2024 – **Graduate Thesis Researcher**, [Columbia University GSAPP](#), New York, NY
- May 2026
- Formulated **S-IRM** and **I-DML** (Invariant DML with ICP pre-filtering) to extract environment-invariant causal features and eliminate spatial wealth confounder bias.
  - Designed an **Adversarial Causal Spatial Autoencoder** (PyTorch) using minimax gradient reversal to disentangle spatial opposition geometry from neighborhood wealth.
- May 2025 – **Research assistant to Dir. Electrical Engineering Zoran Kostic**, [Columbia University](#), New York, NY
- Aug. 2025
- Math benchmarking and finetuning of Deepseek's GRPO algorithm
- Jan. 2025 – **Research assistant to Dir. Financial Engineering Ali Hirs** (**Industry-sponsored research**), [Columbia University](#), New York, NY
- Oct. 2025
- Developed a multi-asset CVAE latent factor model with Skew-T Mixture priors; achieved 85%  $R^2$  vs. commercial SaaS (75%) and Fama-French (58%) on backtested holdout.
  - Derived a tractable, axiom-compliant per-feature SHAP algorithm using Hessian-based recursive clustering to provide stable temporal regime-dependent attributions for deep learning models.
- July 2019 – **Summer research intern to Pres. AAI Yolanda Gil**, [University of Southern California](#), Los Angeles, CA
- Aug. 2019
- Integrated hydrological models into the [MINT Platform](#), competing year-long with MITRE to provide DARPA the "Airflow for science"

## Teaching

- Jan. 2018 – **Co-instructor**, [The University of Texas at Austin](#), Austin, TX
- Dec. 2020
- Helped design and teach graduate-level courses: *Machine Learning for the Geosciences* for Petrobras engineers, *Intelligent Systems for the Geosciences*, and *Scientific Computation*.

## Select Publications

- 2026 **Hardesty Lewis, D.**, "Predicting Formal Protest Petitions Against Housing Development: Evidence from Austin, Texas". M.S. Thesis, Columbia University GSAPP, 2026. [github.com/dhardestylewis/thesis](https://github.com/dhardestylewis/thesis)
- 2021 Gil, Y., et al. (incl. **D. Hardesty Lewis**), "Artificial Intelligence for Modeling Complex Systems: Taming the Complexity of Expert Models to Improve Decision Making". *ACM TUS*, 2021. DOI: [10.1145/3453172](https://doi.org/10.1145/3453172)
- 2019 Garijo, D., et al. (incl. **D. Hardesty Lewis**), "An intelligent interface for integrating climate, hydrology, agriculture, and socioeconomic models". *ACM IUI '19*, 2019. DOI: [10.1145/3308557.3308711](https://doi.org/10.1145/3308557.3308711)

## Education

- 2026 **M.S., Urban Planning**, [Columbia University](#), New York City, NY
- 2017 **B.S., Pure Mathematics**, [University of Texas](#), Austin, TX

## Key Projects

- [Predicting Zoning Opposition](#) Developed the first property- and owner-level causal ML model of NIMBYism: Double Machine Learning (DML) over a 193k case-period panel to isolate the spatial tax of formal protest petitions on allowable building height.
- M&A predictor Built M&A prediction model (Oct.–Dec. 2025) with Prof. Eric Talley on 1.5M Compustat firm-quarters; long-only portfolio beat S&P 500 by 2%/yr (2004–2020), peaking at 6%/yr.

## Activities

- Sailing Volunteer Atlantic SAR of the [Cheeki Rafiki](#) Bikepacking US West & East Coasts

## Languages

- Spanish Native Proficiency French Professional Working Proficiency